Predictive Modeling of Social Media Engagement Patterns on Personality Development and Identity Formation Among Teens and Young Adults

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***Abstract*—In this research, we will bring up the topic of identity related to psychological issues among young adults based on social media habits. Using 196 responses that we obtain from an online questionnaire that we made, we analyze information on social media habits such as screen time, frequency of posting, passive content consumption, and interaction behaviors. We use a Random Forest machine learning model to predict psychological risk based on these social media habits . The model that we used achieved a predictive accuracy of 78% using cross validations. The result shows that passive and active social media engagement is usually related to higher levels of identity issues. These results also point out the potential of using online behavior and machine learning to determine psychological risks, helping people better understand their behavior on social media and build healthier habits**

Keywords—Social Media, Machine Learning, Psychological Vulnerability, Identity Issues, Young Adults.

I. Introduction

Social media is now a big part of people's lives. It has an impact on how young adults communicate, voice their opinion, and how they interact with others. Many social media platforms such as Instagram, TikTok and X give people opportunities to interact and connect easily from anywhere. However, these opportunities also raise concern about how social media may affect the mental health and identity development among young adults [1], [2], [4]. During young adulthood, many people are still trying to understand themselves and build their own identity, but social media makes this process more difficult [3], [11].

Many people, especially young adults, spend a large amount of time on social media, which may lower their confidence and affect how they see themselves. Online content also pressures people to appear as if they have a perfect life in order to gain attention and validation online. [9], [12], [15]. Many studies have discussed the effects of social media on young adults, but it's still unclear how different social media habits can affect the development of young adults identity. Studies that were previously used such as surveys and correlation analysis don't fully explain online behavior or predict mental health problems. [6], [14], [20].

Because of limitations of previous studies, researchers can now use machine learning to better understand social media behavior and analyze its effects on young adults. Social media behavior can be identified using data such as screen time, online activity, and posting frequency. [21], [22]. This study uses machine learning to study how social media behavior may raise mental health issues among young adults [24], [26]. This research studies different social media habits among young adults to understand how online behavior may affect mental health and identity development. Machine learning is also used to help analyze on how social media habits may affect young adults [30].

II. Literature review

Social media, mental health, and machine learning are becoming popular topics for researchers to study. This section will review previous studies about online behavior, identity development, and mental health.

A. Social Media and Identity Formation

Social media has changed how people build their identity and interact with others. It is often used to express themselves and shape their identity online [1]. Social media has become one of the things that influence identity development among young adults, especially in urban communities [2]. Studies show that social media can affect identity development through screen time and also through how people behave and express themselves online[3], but social media can also affect how people view themselves and develop their own identity over time [4]. Identity development is also influenced by other factors such as gender and family relationships [5].The topic of identity development still remains important in studies about young adults [6].

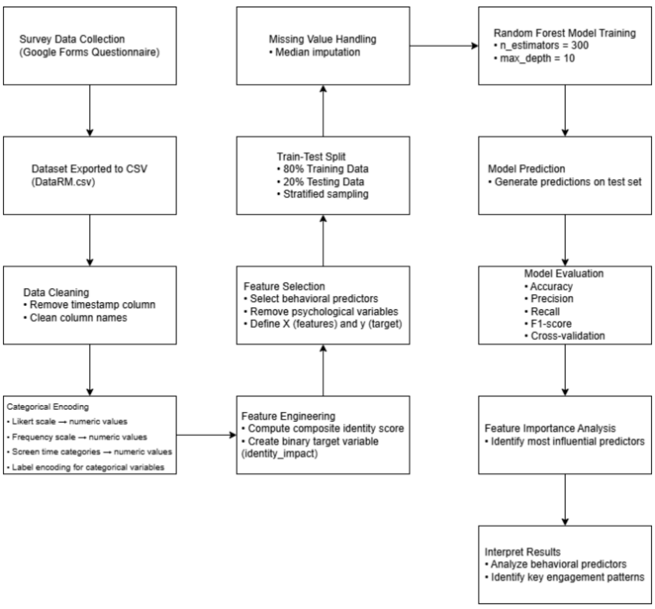
B. Psychological Well-being and Personality Development

Social media can affect mental health and personality in different ways [7]. Using social media too much may increase stress, anxiety, and depression [8], [11]. Social media addiction and loneliness are also still common among university students, which can make it harder for them to understand themselves [12], [13]. Social media can affect how people see their bodies and could lead to unhealthy eating behavior [14] and affect how people feel about their looks [15]. These effects can also influence behavior and personality, so researchers believe that schools should help students create positive behavior and reduce the negative effects of social media [16], [17]. Understanding of social media literacy can help people use social media more carefully and avoid its negative effects [18].

C. Predictive Modeling and Behavioral Analytics

Machine learning is now being used by researchers more often to study and predict the risk of social media use [19]. Researchers also use different types of digital data to help predict mental health issues and online behavior [20]. The way people behave on social media can also be influenced by their personality [21], [22]. Researchers can also use online behavior patterns to better understand mental health problems and identity development [23]. Researchers are improving machine learning models to better understand mental health issues through online behavior [24]. Machine learning is also used to study how people interact on social media and face changes in their lives [25], [26].

III. Methodology



A. Research Method

This project applies quantitative predictive modeling techniques to see if social media behavior predicts identity-based psychological vulnerability in young adults. Instead of performing interviews or experiments in a lab, this study gathered data via a Google Forms survey which we then analyze via machine learning.

The selected model was a Random Forest classifier. Survey data will contain both categorical and numeric features so Random Forest is applicable in this instance. Random forest typically has higher accuracy than one decision tree due to averaging many trees, and does not require much feature engineering in cases of non-linear data.

B. Dataset and Data Collection

We collected data via an anonymous online survey which was hosted and disseminated through social media platforms. Respondents included young adults ranging anywhere from age 14-25. We allowed the survey to collect as many answers until we reached our sample size; 197 total responses were collected. After cleaning our data we had 195 usable responses for our final data set due to one participant not answering all questions. The survey included three sections: the first being general demographic information (age range, sex, and current student/employment status). Section two asked about social media usage including hours spent on social media per day, frequency of posting, engaging with other's posts, and scrolling. We asked additional questions about whether or not they look at their phone when they wake up or before they go to bed. The third section included our five-point Likert scale to gauge agreement with questions about self-image, emotions, and identity formation. This section operationalized our Vulnerability variable.

C. Data Preprocessing

Prior to model training, a few preprocessing steps took place in Python. Metadata columns created automatically by Google Forms (like time stamps) were removed because they aren't useful for predictions.

We transformed answers from the 5 point likert scale to numbers (1 through 5), with the smaller numbers corresponding to negative responses and larger numbers to positive responses. We also dummy encoded other categorical columns (such as gender and platform of choice) in order to run our models. We also examined missing data within our dataset and found that it was quite small. We did impute the few missing values with median responses to maintain consistency. We decided to use the median instead of mean to prevent outliers in our survey from impacting this central value.

D. Target Variable Construction

Psychological vulnerability with respect to identity cannot be reliably measured with just one question. Instead we created a score that represents the mean of how respondents answered across questionnaire questions specific to emotion and identity vulnerability.

After determining the mean score for each individual, we divided our sample into two groups: "low vulnerability" and "high vulnerability". By converting our outcome to a binary classification task we were able to more easily interpret the performance of our Random Forest classifier and maintain the major distinctions between vulnerability in our sample.

E. Model Training and Evaluation

The response variable fed into the Random Forest algorithm was vulnerability score, and the predictor features consisted of all the engineered social media engagement variables from above (screen time, passive consumption, waking/sleeping habits, etc.).

An 80:20 train/test-split was used for modeling. Stratification was used on the data split to ensure that the ratio between the two vulnerability classes was equal in training and testing data. Accuracy, precision, recall, and F1-score were used to determine model performance.

Lastly, 5-fold cross validation was used to determine whether or not our model was overfit and how well it generalized to new data. Each of the cross validation scores were relatively close together which means our model is not too overfit. Our overall accuracy on the final Random Forest model was ~78%. Although not 100%, this is still relatively high and indicates that patterns of social media use are able to provide signals that can help detect higher levels of psychological vulnerability in youth.

IV. RESULTS AND DISCUSSION

We evaluated our model using an 80/20 stratified train-test split and cross-validation, we also measured the model performance using accuracy, precision, recall, and F1-score. Our Random Forest model approached an accuracy of approximately 78%, these results indicate that the way young adults engage with social media shows some behavioral patterns connected with their identity-related psychological vulnerability.

We also extracted the feature importance scores by using the Random Forest model to understand more about which behavioral factors contributed the most to the final prediction. This helped us identify which variables had the strongest influence on the model's final prediction. As shown in *Table 1.1*, the feature importance scores indicate that behavioral patterns of social media engagement are among the strongest predictors of identity-related psychological vulnerability based on the dataset.

Based on the scores, active content posting is the most dominant predictor, surpassing other social media engagement activities such as commenting, messaging, and reacting to other users' posts. In addition, passive content consumption also played a significant role in shaping the prediction output. In conclusion, both active interaction and passive browsing habits on social media contributed to the model's prediction results, showing that online behavior can reflect certain psychological vulnerability patterns.

Beyond that, we found that variables such as favorite social media, overall time spent on social media daily, as well as using social media late at night or early in the morning had some impact on our predictions as well. In comparison, variables containing demographic information such as age ranges, gender, and employment statuses were considered to have very low importance. Therefore, we can conclude that behavior on social media was significantly more valuable in predicting identity-related psychological vulnerability.

While these risk factors were predictive, our model does not establish causality, it only picks up associations. Therefore our findings should be considered associative determinants of behavior that contribute psychological vulnerability.

TABLE I

No.	Results Table	
	Feature	***Importance***
1.	Active Content Posting (Active Usage)	0.213
2.	Commenting / Messaging / Reactions	0.184
3.	Passive Content Consumption	0.122
4.	Primary Social Media Platform	0.109
5.	Daily Social Media Usage Time	0.096
6.	Checking Social Media After Waking / Before Sleeping	0.096
7.	Employment Status	0.072
8.	Age Group	0.069
9.	Gender	0.039

V. CONCLUSION

In this study, we used Machine Learning methods to predict identity-related psychological vulnerability among young adults based on their social media engagement patterns. We collected the data through an online questionnaire. On the questionnaire we use several behavior factors such as screen time, posting frequency, interaction habits, and preferred social media platforms as predictor variables.

We also implemented a Random Forest classification model to analyze the correlation between social media engagement and psychological vulnerability. Based on our evaluation results, our model achieved an accuracy of around 78% using cross-validation. From this result, it shows that social media activity may contain useful behavioral patterns related to identity formation and psychological vulnerability.

Based on the feature importance analysis, active social media engagement especially active content posting and interacting with other users through comments or reactions were the strongest impact on the prediction results. In addition, passive content consumption and daily social media usage also contributed to our model's prediction. And demographic variables such as age group, gender, and employment status had relatively little influence compared to social media behavioral factors.

From the results, we found that social media behavior may be related to psychological vulnerability. However, this model only shows connections between variables and cannot prove direct causes. In future research, larger datasets and additional behavioral factors could be used to improve the prediction results and compare different machine learning algorithms.

VI. OPEN DATA

We collected the data for this study using an online questionnaire that we distributed through Google Form. The processed dataset and analysis code are available in the following repository:

https://drive.google.com/file/d/14gP7lx3hZLzumbX4lXSwCBqHVa8aQ-8 R/view?usp=sharing

We are providing open access to the dataset that we collected to ensure transparency and allow other researchers to reproduce the results of this study.

VII. AI ASSISTANCE DISCLOSURE

Artificial intelligence tools were used during the preparation of this research to help find related literature, improve the wording of survey questions, and assist with paraphrasing and language refinement in the manuscript.

All research design, data collection, machine learning model development, data analysis, and interpretation of results were completed by the authors. All AI-generated suggestions were reviewed and checked by the authors to ensure the information was accurate and appropriate for this study.

VIII. AUTHOR CONTRIBUTORSHIP

Nabil Muhammad Ghifari was responsible for the early stages of the research, including reviewing related literature and finding relevant studies. He also helped collect initial data, organize references, and write the abstract. Willian Yehezkiel Alvin worked on the technical part of the project, including developing the machine learning model, preprocessing the data, and analyzing the results. Meanwhile, Raymond Timothy helped with data collection and project documentation, while also assisting with reference organization and writing the manuscript.

All authors collaborated on the final review and gave their approval for this version of the paper.

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